

Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: WXIMPACTBENCH | Context: Argentine Government Disaster Reporting Reliability — Policy Staff

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark corpus is exclusively English-language historical Canadian newspaper text from La Presse, La Patrie, and the Montreal Gazette [Q97]. No Spanish-language, Argentine, or Latin American source material is represented, and no Argentine institutional actors (SINAGIR, SMN, AFE), geographic references, or local disaster framing appear. The deployment explicitly handles Rioplatense Spanish across formal press, social media, and informal community channels — none of which are in the benchmark. The paper itself acknowledges 'potential biases in underrepresented historical events and linguistic variations' [Q90] and that 'the interpretation of weather-related disruptions in historical newspapers might be influenced by demographic and contextual factors' [Q96]. Empirically, the dataset contains some non-Canadian wire-service content (China, Mediterranean, Indonesia) [DATASET-D18, D19, D21], which marginally broadens geographic scope but does not include South America or Spanish-language content. Web search confirmed no Argentine Spanish-language disaster NLP benchmark exists [WEB-14, WEB-15] to supplement this gap. Given that input_content is a HIGH-priority dimension per elicitation, this represents a major validity violation.

ID	Evidence
Q90	"Although WXIMPACTBENCH provides valuable insights (e.g., exhibit the strengths and weaknesses of various society impact understanding) about evaluating LLMs on disruptive weather, it may have potential biases in underrepresented historical events and linguistic variations." (p.9)
Q96	"The interpretation of weather-related disruptions in historical newspapers might be influenced by demographic and contextual factors, which is similar to other text datasets generated through crowd-sourcing with inherent challenges in ensuring that dataset labels are fully representative of diverse societal perspectives (Talat et al., 2022)." (p.9)
Q97	"Our primary data source is a corpus of three digitized newspapers (La Presse, La Patrie and Montreal Gazette), obtained through collaboration with the McGill University Library and Archives and the Bibliothèque nationale du Québec." (p.9)
D18	more than 100,000 homes damaged and crops on about 175,000 hectares of farmland destroyed... death toll in Chongqing in China's southwest rose to 42 people and 12 missing from torrential downpours — 2007 global flood roundup (England, China, Pakistan) — non-Canadian events used as positive examples
D19	A line of freak thunderstorms rumbling across the northeastern Mediterranean coast from Spain and southern France to the principality of Monaco brought two days of disastrous flash flooding... Raging waters uprooted trees and swept away cars as river levels rose by as much as 3 metres — Mediterranean flooding; ecological and political labels assigned; non-Canadian geography
D21	hundreds of houses, schools and other buildings sustained damage when one of the twin peaks of the volcano exploded. At least 20 people sustained minor injuries... The most powerful earthquake to strike western Washington state in 30 years injured four people — 1999 global disaster roundup (Indonesia, Washington state, NZ, Romania); labeled positive
WEB-14	No complementary Argentine Spanish-language benchmark exists for cross-validation (https://crisisnlp.qcri.org/humaid_dataset)
WEB-15	https://ojs.aaai.org/index.php/ICWSM/article/download/18116/17919/21611

Strengths

- Some non-Canadian wire-service content (China, Pakistan, Mediterranean, Indonesia) appears in positive examples [DATASET-D18, D19, D21], indicating impact categories are applied beyond strictly Canadian events
- Two temporal periods covered to introduce linguistic variation [Q13]
- Authors explicitly acknowledge potential biases in underrepresented events [Q90]

ID	Evidence
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Q13	"We extract a sample of articles from two time periods, which cover linguistic and social changes across different eras and increase linguistic complexity due to the different descriptions of weather events in different times (Campbell, 2013)." (p.2)
Q90	"Although WXIMPACTBENCH provides valuable insights (e.g., exhibit the strengths and weaknesses of various society impact understanding) about evaluating LLMs on disruptive weather, it may have potential biases in underrepresented historical events and linguistic variations." (p.9)
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Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Argentine deployment requires Rioplatense Spanish proficiency, recognition of Argentine institutional actors (SINAGIR, AFE, SMN, Prefectura Naval, provincial civil defense), and familiarity with local disaster framing including villa miseria vulnerability — none of which appears in benchmark inputs [Q97].

IC-2: Cultural alignment is poor: benchmark content is North American historical journalism [Q97]; elicitation flags that 'informal or social media data carries meaning and communication patterns that are likely not represented in the Canadian-sourced benchmark' [A2].

IC-3: Inputs require Western-specific knowledge: 19th-century Canadian institutional context (Q12, Q14), Montreal civic geography [DATASET-D16], and English-language journalism conventions. None transfers to Spanish-language Argentine social media or AMBA disaster reporting.

IC-4: INSUFFICIENT DOCUMENTATION on regional annotator recruitment for Argentine context; no Argentine annotators were involved per Q110 (annotators specialize in regional Canadian historical climate).

IC-5: Major content issues: zero Spanish-language content; zero Argentine geography; zero villa miseria framing; zero social media content; zero institutional actors relevant to AFE/SINAGIR. Confirmed via web search — no complementary Argentine Spanish disaster benchmark exists [WEB-14, search gap_id 2 and 7].

ID	Evidence
Q12	"We design a four-stage data construction pipeline that begins with a disruptive weather impact dataset in which we correct OCR errors in digitalized newspaper article extraction." (p.2)
Q14	"Historical newspapers often employed more descriptive and elaborate narratives compared to modern reporting styles (Bingham, 2010). These narratives frequently included outdated terminology, spelling variations, and evolving writing conventions (Campbell, 2013)." (p.2)
Q90	"Although WXIMPACTBENCH provides valuable insights (e.g., exhibit the strengths and weaknesses of various society impact understanding) about evaluating LLMs on disruptive weather, it may have potential biases in underrepresented historical events and linguistic variations." (p.9)

Q97	"Our primary data source is a corpus of three digitized newspapers (La Presse, La Patrie and Montreal Gazette), obtained through collaboration with the McGill University Library and Archives and the Bibliothèque nationale du Québec." (p.9)
Q110	"The annotation process was conducted by members of a research group specializing in uncovering the history of a region's climate change through the regional historical weather records." (p.15)
D16	Aid Laurent said that there were mountains of snow all over the city. Already \$30,000 has been expended in removing it... Seven hundred and fifty men and 450 carters were employed... — 1887 Montreal snow removal debate in city council; all three labeled labels are plausible
WEB-1	SINAGIR/AFE Argentine institutional framework operates with different political accountability structure than Canadian framework (https://www.argentina.gob.ar/sinagir/institucional)
WEB-5	98.4% Argentine mobile broadband indicates short-form mobile content dominant in deployment (https://datareportal.com/reports/digital-2026-argentina)
WEB-14	No complementary Argentine Spanish-language benchmark exists for cross-validation (https://crisisnlp.qcri.org/humaid_dataset)

Information Gaps

- Whether HumAID/CrisisBench English-language tweets could serve as partial supplements after cross-lingual transfer validation [WEB-14]
- Empirical performance of LLMs on Rioplatense Spanish disaster content (no Argentine benchmark exists for direct measurement)

Requires Expert Verification

- Whether formal Argentine press content (Clarín, Infobae, La Nación) closely follows transferable journalistic patterns as A2 hypothesizes
- Construction of an Argentine Spanish gold-standard test set for cross-lingual validation

Remediation

Gap 1: Zero Spanish-language, Argentine, or social media content in benchmark corpus [Q97, WEB-14]

Recommendation: Construct an Argentine Spanish disaster impact test set drawing from Clarín, Infobae, La Nación, government press releases, and Twitter/X content covering AMBA flooding, sudestada, Pampa drought, and Patagonian wind events. Use Argentine emergency management experts as annotators. Use HumAID/CrisisBench as methodological reference (not direct content reference) given the English-language gap.

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Q97	"Our primary data source is a corpus of three digitized newspapers (La Presse, La Patrie and Montreal Gazette), obtained through collaboration with the McGill University Library and Archives and the Bibliothèque nationale du Québec." (p.9)
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